

## Expected Deliverables

### 1. Machine Learning Model:

- **Model Selection:** The pipeline utilizes a Random Forest Regressor. This ensemble model was chosen for its ability to handle non-linear relationships and its resistance to overfitting compared to single decision trees.

Evaluation Metrics (Performance on Unseen Test Data): The model was tested on a 20% hold-out validation set.

- **R-squared ( $R^2$ ) Score: 0.3707**

*Interpretation:* The model explains approximately **37%** of the variance in future liquidity. This indicates a moderate predictive capability, which is typical for highly volatile financial markets like cryptocurrency.

- **Root Mean Squared Error (RMSE): 1.2683**

*Interpretation:* On average, the model's predictions deviate from the actual scaled liquidity values by 1.26 units.

### Key Insights:

- **Feature Importance:** The model identified **Current 24h Volume** (57% importance) and **Market Cap** (37% importance) as the strongest predictors of future liquidity. Price fluctuation (1h/24h/7d changes) had minimal impact on the prediction.

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## 2. Data Processing & Feature Engineering

### Data Cleaning & Preparation:

- **Imputation:** Missing values in the raw dataset were filled using the mean value specific to each cryptocurrency symbol. Remaining gaps were filled with the overall column mean.
- **Deduplication:** Duplicate entries were removed, resulting in a clean dataset of **1,000 rows** (representing 500 coins over 2 days).

- **Scaling:** Numerical features (`price`, `1h`, `24h`, `7d`, `24h_volume`, `mkt_cap`) were normalized using **StandardScaler** to ensure they have a mean of 0 and a standard deviation of 1. This prevents features with large raw numbers (like Market Cap) from biasing the model.

### Feature Engineering Strategy:

- **Time-Series Structuring:** The data was sorted by Coin Symbol and Date.
- **Target Variable Definition:** A new column, `target_liquidity_volume`, was created by **shifting the data by -1 day**.
  - *Logic:* The model uses *today's* metrics (Price, Market Cap, Volume) to predict *tomorrow's* Volume (Liquidity).
- **Dataset Trimming:** The last day of data for each coin was dropped because it lacked a future target value to predict, resulting in a final training shape of **497 rows**.